

Midterm 1

EECS 245, Spring 2026 at the University of Michigan

Name: _____

username: _____

UMID: _____

Room: ☐ 1690 BBB (1-3PM) ☐ 1690 BBB (extended time)

Instructions

- This exam consists of 9 problems, worth 100 points, spread across 12 pages (6 sheets of paper).
- You have 120 minutes to complete this exam, unless you have extended-time accommodations through SSD.
- Write your username in the top right corner of each page.
- For free response problems, **you must show all of your work**, and circle your final answer. We will not grade work that appears elsewhere, and you may lose points if your work is not shown.
- For multiple choice problems, completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.
 - ☐ A bubble means that you should only select one choice.
 - ☐ A square box means that you should select all that apply.
- You may refer to **one double-sided 8.5x11" handwritten notes sheet**. Other than that, you may not refer to any other resources or technology during the exam (no phones, watches, or calculators).

You are to abide by the University of Michigan/Engineering Honor Code. To receive a grade, please sign below to signify that you have kept the Honor Code pledge.

I have neither given nor received aid on this exam, nor have I concealed any violations of the Honor Code.

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Tip: Skim through the entire exam before starting to work on it.

Problem 1 (16 pts)

Suppose we'd like to find the optimal parameter, w^* , for the constant model $h(x_i) = w$, using the following dataset of $n = 4$ values, y_1, y_2, y_3, y_4 :

0, 2, 4, ~~20~~

$1 \leq w \leq 3$

- a) (3 pts) First, suppose we find the optimal parameter by minimizing mean squared error, $R_{sq}(w)$. Which value of w minimizes $R_{sq}(w)$? Give your answer as a number with no variables.

minimizer of $R_{sq}(w) =$

6.5

$$\frac{0+2+4+20}{4} = \frac{1}{2} + 1 + 5 = 6 + \frac{1}{2}$$

Now, consider the **clipped** loss function, defined below

$$L_{clip}(y_i, h(x_i)) = \min\{(y_i - h(x_i))^2, 9\}$$

For example, $L_{clip}(10, 5) = 9$ and $L_{clip}(5, 3) = 4$.

Let $R_{clip}(w)$ be the average clipped loss for the constant model and this dataset.

- b) (3 pts) State one value of w where the derivative of $R_{clip}(w)$ is not defined.

one value of w where the derivative of $R_{clip}(w)$ is not defined =

17

any $w = y_i \pm 3$ works!

23, 7, 1, 5, -1, 3, -3

- c) (3 pts) Suppose we restrict w to the interval $1 \leq w \leq 3$. Among all values of w in this interval, which value minimizes $R_{clip}(w)$? Give your answer as a number with no variables.

minimizer of $R_{clip}(w)$ within the interval $[1, 3] =$

2

- d) (3 pts) Now suppose there are no restrictions on w . Among all possible values of w , which value minimizes $R_{clip}(w)$? Give your answer as a number with no variables.

minimizer of $R_{clip}(w) =$

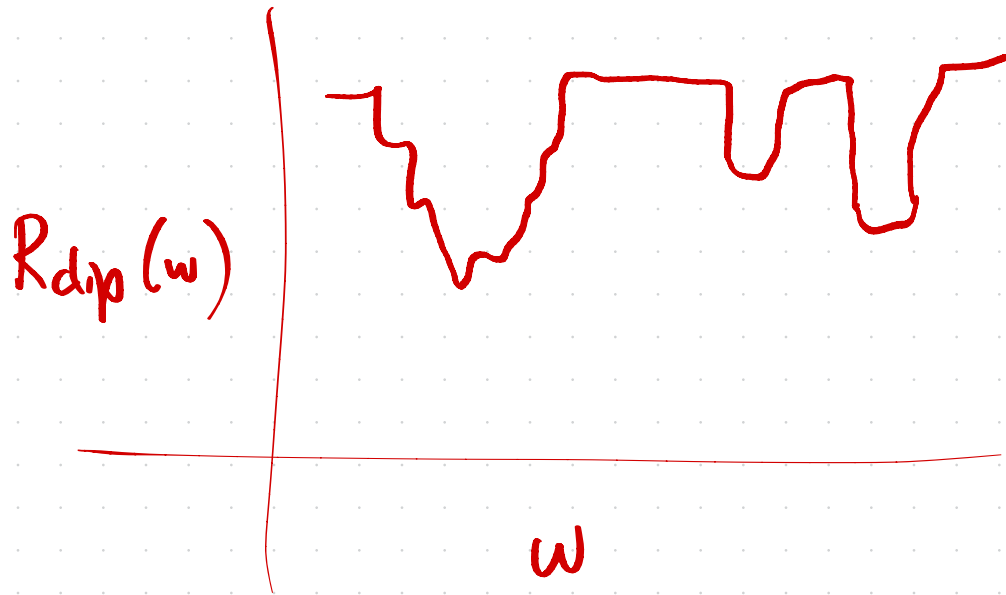
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- e) (4 pts) State one pro and one con of using clipped loss instead of squared loss to find optimal model parameters.

pro: reduces impact of outliers ✓
con: $R_{clip}(w)$ not always differentiable:
harder to find w^*

Graph of $R_{clip}(w)$



$$R_{clip}(6.5) = \frac{9 + 9 + (6.5 - 4)^2 + 9}{4}$$

$(6.5 - 0)^2 > 9$,
so return 9

0, 2, 4, 20

$$R_{clip}(2) = \frac{(2 - 0)^2 + (2 - 2)^2 + (2 - 4)^2 + 9}{4}$$

Problem 2 (10 pts)

We will continue to use the constant model, $h(x_i) = w$, and the same dataset of $n = 4$ values as in Problem 1:

$$0, \quad 2, \quad 4, \quad 20$$

Instead of the clipped loss function, consider the **weighted absolute** loss function, defined below.

$$L_{\text{WA}}(y_i, h(x_i)) = \begin{cases} \beta(y_i - h(x_i)), & h(x_i) < y_i \\ h(x_i) - y_i, & h(x_i) \geq y_i \end{cases}$$

where β is a positive integer. Let $R_{\text{WA}}(w)$ be the average weighted absolute loss for the constant model and this dataset.

The slope of $R_{\text{WA}}(w)$ at w , for any value of w not equal to one of the y_i values, is

$$\text{slope of } R_{\text{WA}}(w) \text{ at } w = \frac{\# \text{ left of } w - \beta(\# \text{ right of } w)}{4}$$

- a) (4 pts) Suppose $\beta = 1$. Which value of w minimizes $R_{\text{WA}}(w)$? Show your work, and write your final answer in the box provided. Your answer should be a number with no variables. If there are multiple possible answers, state just one.

minimizer of $R_{\text{WA}}(w) =$

- b) (6 pts) Now suppose $\beta = 2$. Which value of w minimizes $R_{\text{WA}}(w)$? Show your work, and write your final answer in the box provided. Your answer should be a number with no variables. If there are multiple possible answers, state just one.

minimizer of $R_{\text{WA}}(w) =$

Problem 3 (14 pts)

Suppose we fit a simple linear regression model **with** an intercept term, $h(x_i) = w_0 + w_1 x_i$, to a dataset of n points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ by minimizing mean squared error. You are given the following information:

- The fit model satisfies $h(-4) = 5$ and $h(8) = 14$.
 - The mean of $y_1, y_2, \dots, y_n$ is $\bar{y} = 2$.
- a) (6 pts) Find $\bar{x}$, the mean of $x_1, x_2, \dots, x_n$. Show your work, and write your final answer in the box provided. Your answer should be a number with no variables. *Hint: What property does the line $h(x_i)$ satisfy?*

$\bar{x} =$

- b) (4 pts) Suppose the correlation coefficient between the x -values and y -values is $r = 1/3$. The standard deviation of y , σ_y , is c times the standard deviation of x , σ_x . In other words,

$$\sigma_y = c\sigma_x$$

What is the value of c ?

- ☐ $1/4$
 ☐ $4/9$
 ☐ $3/4$
 ☐ $9/4$
 ☐ 3
 ☐ 4

- c) (4 pts) Let $e_i = y_i - h(x_i)$ be the fit model's error for the i th point. Note that e_i may either be positive or negative. Which of the following statements are **guaranteed** to be true? **Select all** that apply.

☐ $\sum_{i=1}^n e_i = 0$

☐ $\sum_{i=1}^n x_i e_i = 0$

☐ $\sum_{i=1}^n y_i e_i = 0$

☐ $\sum_{i=1}^n e_i (x_i - \bar{x}) = 0$

Problem 4 (8 pts)

Let $\vec{u}, \vec{v} \in \mathbb{R}^n$ be vectors satisfying

$$\|\vec{v}\| = 5, \quad \|\vec{u} + \vec{v}\| = 10, \quad \|\vec{u} - \vec{v}\| = 6$$

Find $\|\vec{u}\|^2$ (**not** $\|\vec{u}\|$). Show your work, and write your final answer in the box provided. Your answer should be a number with no variables.

$$\|\vec{u}\|^2 = \boxed{}$$

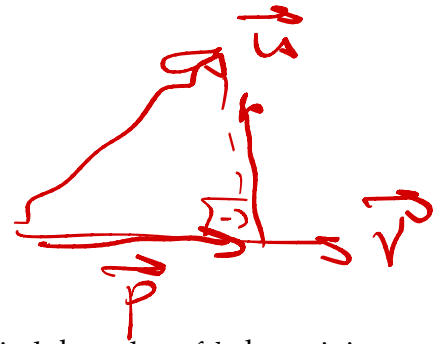
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Problem 5 (13 pts)

Suppose $\vec{u}, \vec{v} \in \mathbb{R}^n$ are non-zero vectors and k is a scalar. Let

$$f(k) = \|\vec{u} - k\vec{v}\|^2 + Ck^2$$

where $C \geq 0$ is a non-negative constant.



- a) (6 pts) In this part only, suppose $C = 0$, $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and $\vec{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Find the value of k that minimizes $f(k)$. Show your work, and write your final answer in the box provided. Your answer should be a number with no variables.

Handwritten work for part a):

$$f(k) = \left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} - k \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\|^2$$

$$= \left\| \begin{bmatrix} 1-3k \\ 2-k \end{bmatrix} \right\|^2$$

$$= \sqrt{(1-3k)^2 + (2-k)^2}$$

$$= (1-3k)^2 + (2-k)^2$$

$$= 10k^2 - 10k + 5$$

$$f'(k) = 20k - 10 = 0$$

$$k^* = 1/2$$

minimizer of $f(k) = \boxed{1/2}$

- b) (4 pts) Note that $f(k)$ almost looks like the squared norm of the vector $\vec{u} - k\vec{v}$, but with an extra term Ck^2 . Let's try and rewrite $f(k)$ so that it is the squared norm of another related vector. Define two new vectors, $\vec{U}, \vec{V} \in \mathbb{R}^{n+1}$ by appending the scalar a to the end of $\vec{u}$ and the scalar b to the end of $\vec{v}$.

Handwritten work for part b):

$$k = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v} + C}$$

$$\vec{U} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \\ a \end{bmatrix}, \quad k\vec{V} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \\ b \end{bmatrix}$$

$$\|\vec{U} - k\vec{V}\|^2 = \sum_i (u_i - kv_i)^2 + (a - kb)^2$$

$$= \sum_i (u_i - kv_i)^2 + (a - kb)^2$$

Select values of a and b so that $f(k) = \|\vec{U} - k\vec{V}\|^2$, for all possible non-negative values of C .

- (i) What is the value of a ? ☒ 0 ☐ C ☐ C² ☐ $\sqrt{C}$
- (ii) What is the value of b ? ☐ 0 ☐ C ☐ C² ☒ $\sqrt{C}$

- c) (3 pts) As C increases, what happens to the value of k that minimizes $f(k)$? Explain your reasoning.

Handwritten work for part c):

$$f(k) = \|\vec{u} - k\vec{v}\|^2 + Ck^2$$

$$k = \frac{(\vec{u} - k\vec{v}) \cdot (\vec{u} - k\vec{v})}{\|\vec{u} - k\vec{v}\|^2} + Ck$$

$$= \frac{\vec{u} \cdot \vec{u} - 2k(\vec{u} \cdot \vec{v}) + k^2(\vec{v} \cdot \vec{v})}{\|\vec{u} - k\vec{v}\|^2} + Ck$$

$$f'(k) = -2(\vec{u} \cdot \vec{v}) + 2k(\vec{v} \cdot \vec{v}) + 2Ck = 0$$

$$k^* = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v} + C}$$

Problem 6 (11 pts)

Suppose $c \in \mathbb{R}$ is a constant and

$$\vec{u} = \begin{bmatrix} 3 \\ 1 \\ c \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 6 \\ c \\ -2 \end{bmatrix}$$

a) (4 pts) Fill in the blanks to complete the sentence:

For all values of c , $\text{span}(\{\vec{u}, \vec{v}\})$ is a __(i)__-dimensional subspace of __(ii)__.

(i):

(ii):

b) (7 pts) Suppose the plane spanned by $\vec{u}$ and $\vec{v}$ is

$$ax + 24y + 3z = 0$$

where a is also a constant. Find the value of c . Show your work in the space provided, and write your final answer in the box provided. Your answer should be a number with no variables.

$c =$

$$a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3 + d\vec{v}_4 = \vec{0}$$

unique name: _____

Problem 7 (10 pts)

Suppose $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4 \in \mathbb{R}^n$ are a **linearly independent** collection of vectors. Define

$$\vec{p} = \vec{v}_1 + \vec{v}_2, \quad \vec{q} = \vec{v}_2 + \vec{v}_3, \quad \vec{r} = \vec{v}_3 + \vec{v}_4, \quad \vec{s} = \vec{v}_4 + \vec{v}_1$$

a) (7 pts) Are $\{\vec{p}, \vec{q}, \vec{r}, \vec{s}\}$ linearly independent?

(i) Select an answer: ☐ Yes ☐ No

(ii) Prove your answer using the formal definition of linear independence. Hint: You did something similar in Homework 4, Problem 6.

$$\begin{aligned} a_1\vec{p} + a_2\vec{q} + a_3\vec{r} + a_4\vec{s} &= \vec{0} \\ \vec{p} + \vec{q} + \vec{r} &= \vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \vec{v}_4 \\ \vec{p} + \vec{q} + \vec{r} - 2\vec{q} &= \vec{v}_1 + \vec{v}_2 + \vec{v}_3 + \vec{v}_4 - 2\vec{v}_2 \\ \vec{p} - \vec{q} + \vec{r} &= \vec{v}_1 + \vec{v}_3 + \vec{v}_4 \\ \vec{p} - \vec{q} + \vec{r} - \vec{s} &= \vec{v}_1 + \vec{v}_3 + \vec{v}_4 - (\vec{v}_4 + \vec{v}_1) \\ &= \vec{0} \end{aligned}$$

b) (3 pts) What is the dimension of $\text{span}(\{\vec{p}, \vec{q}, \vec{r}, \vec{s}\})$? Give your answer as a number with no variables.

$$\dim(\text{span}(\{\vec{p}, \vec{q}, \vec{r}, \vec{s}\})) =$$

$$\begin{aligned} a_1\vec{v}_1 + a_1\vec{v}_2 + a_2\vec{v}_3 + a_3\vec{v}_4 &= \vec{0} \\ \downarrow + a_2\vec{v}_2 + a_3\vec{v}_3 & \\ a_1\vec{v}_1 + (a_1 + a_2)\vec{v}_2 + (a_2 + a_3)\vec{v}_3 & \\ + a_3\vec{v}_4 &= \vec{0} \end{aligned}$$

$$\vec{p}, \vec{q}, \vec{r}$$

$$a_1\vec{p} + a_2\vec{q} + a_3\vec{r} = \vec{0}$$

$$a_1\vec{v}_1 + (a_1 + a_2)\vec{v}_2 + (a_2 + a_3)\vec{v}_3 + a_3\vec{v}_4 = \vec{0}$$

$$a_1(\vec{v}_1 + \vec{v}_2) + a_2(\vec{v}_2 + \vec{v}_3) + a_3(\vec{v}_3 + \vec{v}_4) = \vec{0}$$

Problem 8 (8 pts)

Suppose S is the subspace of $\mathbb{R}^4$ defined by

$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 : x_1 - x_2 + x_3 - x_4 = 0 \right\}$$

Which of the following sets is a basis for S ? **Select all** that apply.

☐ $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

☐ $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

☐ $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

☐ $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$

max 1
min -1

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Problem 9 (10 pts)

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$$

- a) (7 pts) Suppose x and y are non-negative numbers. Using the Cauchy-Schwarz inequality, prove that

$$\frac{(x+y)^2}{2} \leq x^2 + y^2$$

Solutions that do not use the Cauchy-Schwarz inequality will not receive credit.

Let's experiment!

$$\vec{u} = \begin{bmatrix} x+y \\ \frac{1}{2}(x+y) \end{bmatrix} \quad \vec{v} = \begin{bmatrix} x+y \\ -\frac{1}{2}(x+y) \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = (x+y)^2 - \frac{1}{4}(x+y)^2 = \frac{3}{4}(x+y)^2$$

$$\vec{u} = \begin{bmatrix} x+y \\ x+y \end{bmatrix} \quad \vec{v} = \begin{bmatrix} x+y \\ -\frac{1}{2}(x+y) \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = (x+y)^2 - \frac{1}{2}(x+y)^2 = \frac{1}{2}(x+y)^2$$

see next page!

$$\|\vec{u}\| = \sqrt{(x+y)^2 + (x+y)^2} = \sqrt{2}(x+y)$$

$$\|\vec{v}\| = \sqrt{(x+y)^2 + \frac{1}{4}(x+y)^2} = \frac{\sqrt{5}}{2}(x+y)$$

- b) (3 pts) Now suppose x , y , and z are non-negative numbers. Which inequality is guaranteed to be true?

☐ $\frac{(x+y+z)^2}{2} \leq x^2 + y^2 + z^2$

☒ $\frac{(x+y+z)^2}{3} \leq x^2 + y^2 + z^2$

☐ $\frac{(x+y+z)^2}{2} \leq x^3 + y^3 + z^3$

☐ $\frac{(x+y+z)^3}{3} \leq x^3 + y^3 + z^3$

☐ None of the above

Using approach from next page:

$$\vec{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$$

$$x+y+z \leq \sqrt{x^2+y^2+z^2} \sqrt{3}$$

$$\frac{(x+y+z)^2}{3} \leq x^2 + y^2 + z^2$$

$$\frac{(x+y)^2}{2} \leq x^2 + y^2 \leftarrow \text{goal}$$

$$\vec{u} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{u} \cdot \vec{v} = x + y$$

$$\|\vec{u}\| = \sqrt{x^2 + y^2}$$

$$\|\vec{v}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$$

$$x + y \leq \sqrt{x^2 + y^2} \sqrt{2}$$

square both sides $\rightarrow$ legal because
both sides ≥ 0 !

$$(x+y)^2 \leq (x^2 + y^2)(2)$$

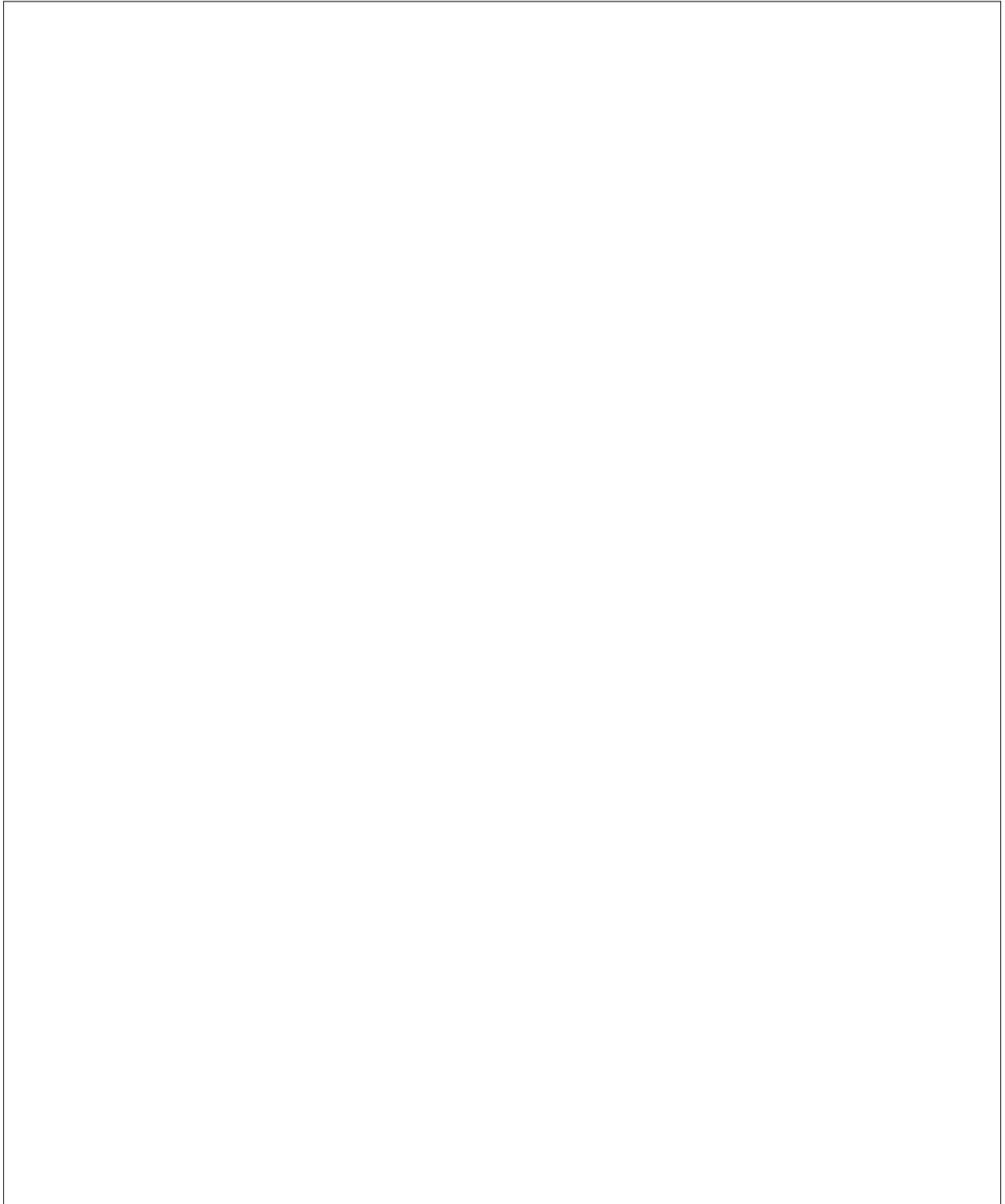
$$\boxed{\frac{(x+y)^2}{2} \leq x^2 + y^2} \quad \checkmark$$

Some other strategies

$$\vec{u} = \begin{bmatrix} a \\ b \end{bmatrix} \quad \vec{v} = \begin{bmatrix} b \\ a \end{bmatrix}$$

$$\vec{u} = \begin{bmatrix} a \\ b \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 1/a \\ 1/b \end{bmatrix}$$

Congrats on finishing Midterm 1! Feel free to draw us a picture about EECS 245 in the box below.

A large, empty rectangular box with a thin black border, intended for a drawing about EECS 245.